03 - Text Baseline

Goal: build a first text model and understand its errors.

1. Largest-class baseline: the lowest score any model must beat.
2. TF-IDF + Linear SVM: a fast, standard text classifier.
3. Examples: what the model looks at for one product.
4. Experiments: text cleaning, binary TF-IDF, number of features.
5. Error analysis and per-category scores, text vs image.

1. Load the Prepared Text

We load the cleaned text with the shared 80/20 train/validation split. The model uses the product title and description joined together.

```
Training products:   67932
Validation products: 16984
```

Analysis

- 67932 training products (80%) and 16984 validation products (20%).
- The split is stratified, so each category has about the same share in both sets.
- All models (text, image, fusion) use this same split.

2. Largest-Class Baseline

This model always predicts the most common category. It ignores the text and gives the lowest score a useful model must beat.

```
Largest class: 2583
Weighted F1:   0.0258
Macro F1:      0.0080
```

Analysis

- Always predicting category 2583 gives a weighted F1 of 0.0258 and a macro F1 of 0.0080.
- With 27 categories, this is almost always wrong.
- Any real model must be far above this.

3. Convert Text into TF-IDF Features

TF-IDF turns each text into numbers, one per word or word pair. We keep the 50000 most frequent ones. The vocabulary is learned on the training set only.

```
Training matrix:   (67932, 50000)
Validation matrix: (16984, 50000)
```

Analysis

- Each product becomes a vector of 50000 values, one per word or word pair.
- Words specific to a category (*playstation*, *sommier*) get a high weight. Common words (*pour*, *avec*) get a low weight.
- Learning the vocabulary on train only avoids leaking validation data.

4. Train the First Text Classifier (Linear SVM)

A linear SVM is a standard choice for TF-IDF features. It trains in seconds.

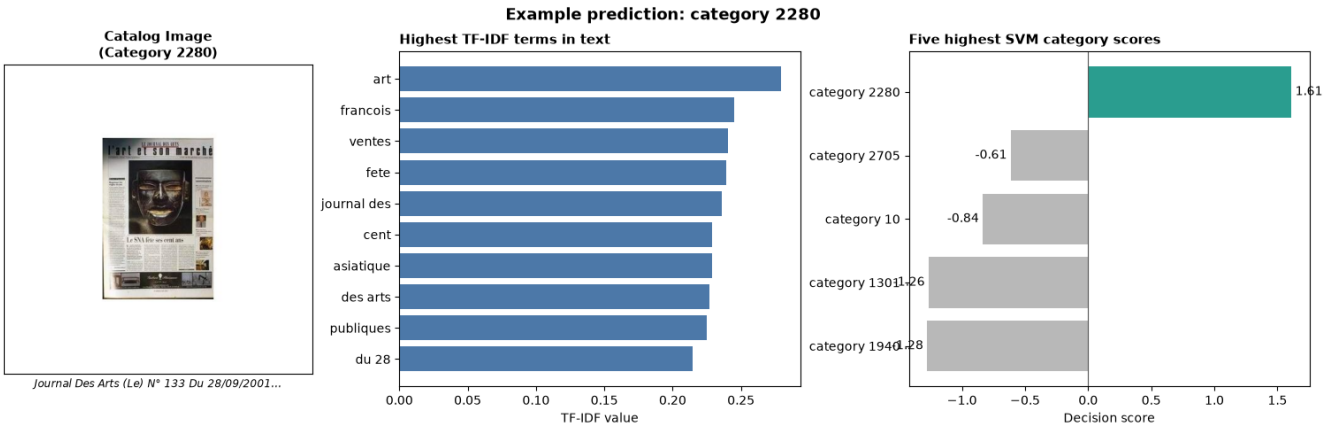
Model	Weighted F1	Macro F1
Largest-class baseline	0.0258	0.0080
TF-IDF + Linear SVM	0.8274	0.8156

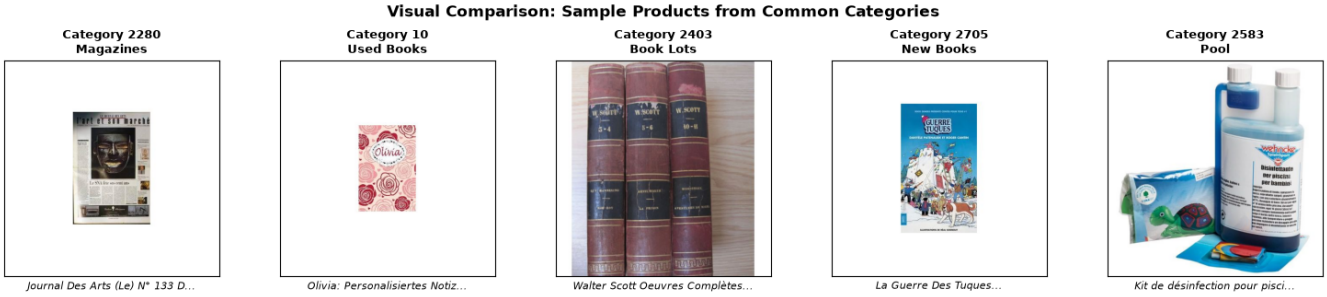
Analysis

- Weighted F1: 0.8274.
- Macro F1: 0.8156. This is close to the weighted score, but it does not mean every category is good. The per-category table below shows some much lower (Used Books 0.53).

5. What the Model Looks At

We take one correctly predicted product and show its image, its highest TF-IDF words and the model's scores for each category. We also show one example image for some common categories.





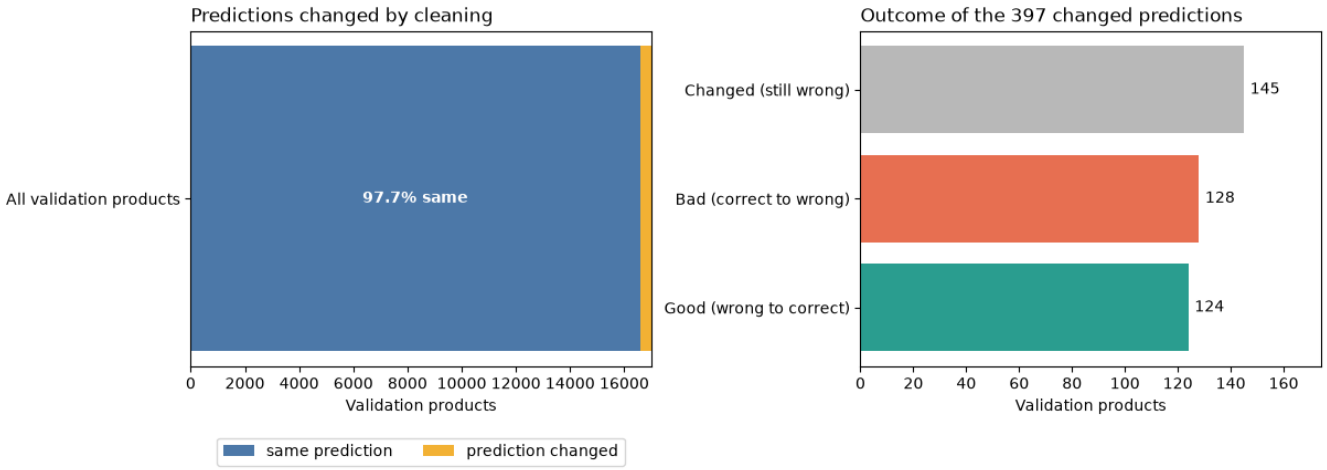
Analysis

- For the art journal (category 2280), words like *art*, *ventes*, *journal des* and *des arts* give 2280 the highest score.
- Book lots (2403), magazines (2280) and used books (10) share words like *édition* and *tome*, but their images can look different. The per-category table below checks if images help.

6. Did the Text Cleaning Help?

We train the same model on the raw text (title + description, without cleaning) and compare.

Text	Weighted F1	Macro F1
raw text	0.8276	0.8167
cleaned text	0.8274	0.8156




```
Products with identical predictions: 97.7%
Net change in correct classifications: -4
```

```
Example 1: True Category 10 (Raw predicted: 10 [Correct] to Clean predicted: 2705
[Wrong])
  RAW:   "L'avant Scene N° 190 Meurtres En Fa Diese -Sait On Jamais "
  CLEAN: "l'avant scene n° 190 meurtres en fa diese -sait on jamais"

Example 2: True Category 1320 (Raw predicted: 1320 [Correct] to Clean predicted:
2060 [Wrong])
  RAW:   'Suspension Décoration Murale Merlin <div class="rakuten_rich_content"
><div> <h3>Suspension décoration mu'
  CLEAN: 'suspension décoration murale merlin suspension décoration murale sur le
thème merlin 100% polyester'

Example 3: True Category 10 (Raw predicted: 10 [Correct] to Clean predicted: 2280
[Wrong])
  RAW:   'Magies Noires Ou Les Magies Qui Font Peur '
  CLEAN: 'magies noires ou les magies qui font peur'
```

Binary TF-IDF

With binary TF-IDF, each word counts once if it appears, however many times it is repeated.

	TF-IDF Mode	Weighted F1	Macro F1
0	Standard (Frequency-weighted)	0.8274	0.8156
1	Binary (Word Presence Only)	0.8313	0.8192

```
Binary TF-IDF recovers 58 of the 128 cases hurt by cleaning (45.3%).
Overall Weighted F1 improves from 0.8274 to 0.8313.
```

Analysis

- Some products were hurt by cleaning because sellers repeat the title inside the description. Repeated words get more weight. For example, *décoration* repeated several times pushes a baby product (1320) to Home Decor (2060).
- With binary TF-IDF, 58 of these 128 errors are fixed, and weighted F1 goes from 0.8274 to 0.8313.
- Binary TF-IDF was chosen by looking at validation errors. The next section checks if the gain is real.

Is the Binary Gain Real?

The gain above comes from one split. We re-split the data with 5 different seeds and train both versions on each.

Seed	Normal TF-IDF	Binary TF-IDF	Gain
0	0.8260	0.8311	0.0051
1	0.8213	0.8254	0.0041
2	0.8242	0.8285	0.0044
3	0.8258	0.8255	-0.0004
4	0.8284	0.8316	0.0033

```
Average gain:        +0.0033
Gain on every seed:  False
Normal F1 spread:    0.0071
```

Analysis

- Binary wins on 4 of 5 seeds. The average gain is 0.33 points.
- The score itself moves by 0.71 points between seeds, twice the gain.
- With 50000 features, the binary gain is too small to be sure. It is tested again in the optimisation notebook (04).

How Many Features Do We Need?

We count how many words and word pairs exist in total. Then we test several limits with 3-fold cross-validation on train only.

```
Total features available: 460326
Dropped by the 50k limit: 410326
```

Feature limit	CV weighted F1
20000	0.8027
50000	0.8164
100000	0.8231
no limit	0.8284

Analysis

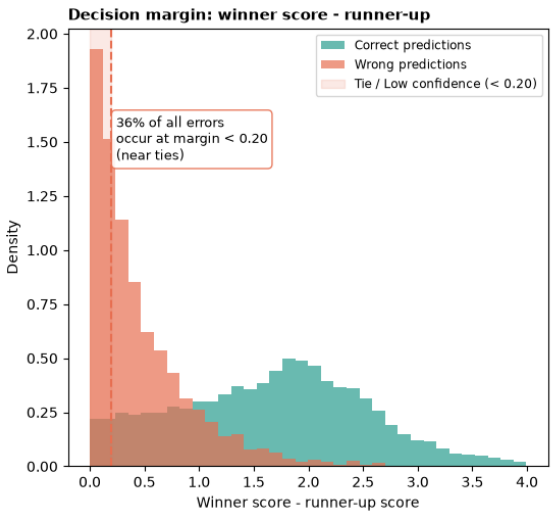
- There are 460326 features in total. The 50000 limit drops 410326 of them (89%).
- CV weighted F1: 20k gives 0.8027, 50k gives 0.8164, 100k gives 0.8231, no limit gives 0.8284.
- No limit beats 50k by 1.2 points, more than the seed noise (0.71). The limit is removed in the optimisation notebook (04).

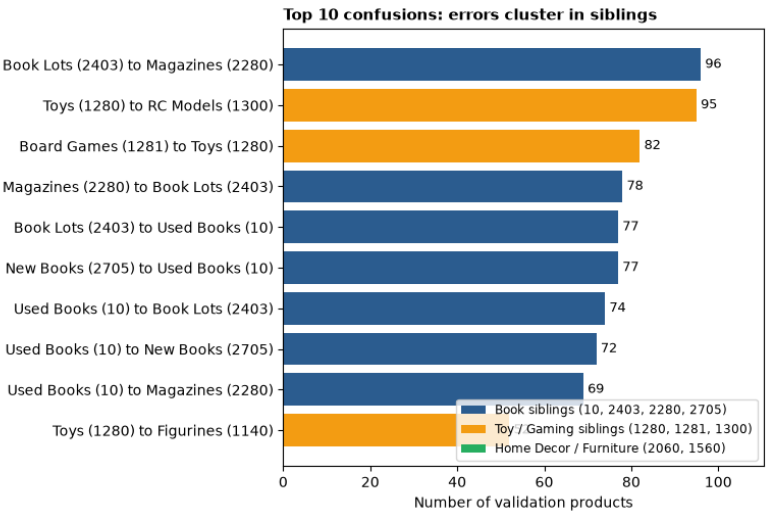
About the 0.8113 benchmark: it is a CNN on titles only, scored on Rakuten's hidden test set. Our model uses title and description and is scored on our own validation split, so the two numbers cannot be compared directly.

7. Error Analysis

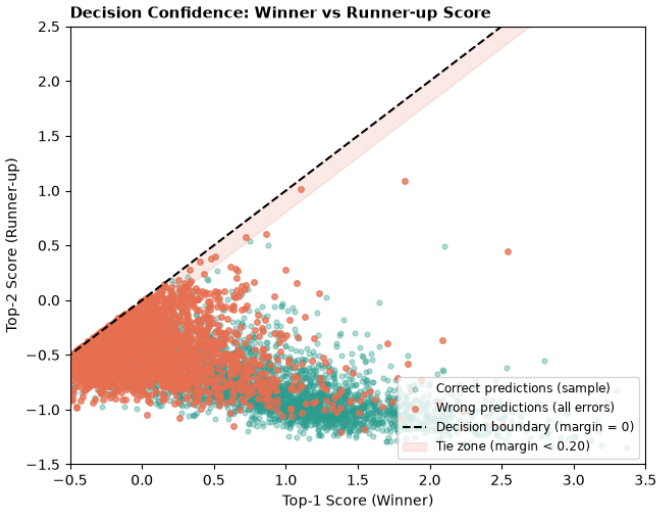
We look at two things:

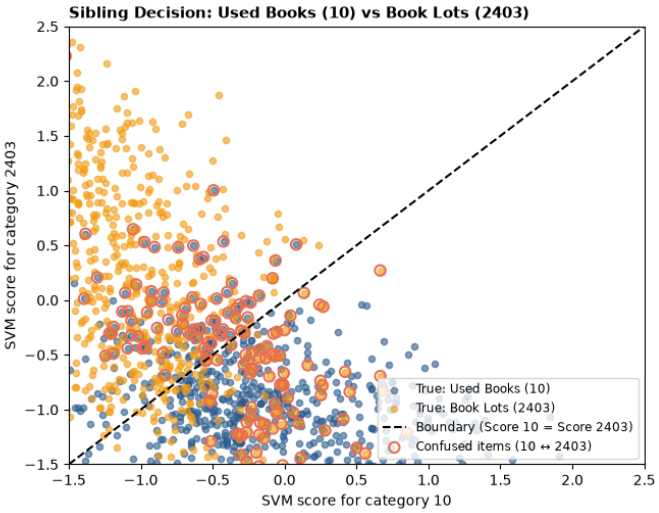
1. Decision margin: the score of the winning category minus the score of the second one. A small margin means the model was unsure.
2. The category pairs the model mixes up most often.






```
Validation errors in low-confidence zone (margin < 0.20): 1036 (35.6%)
```







Analysis

- Errors happen when the model is unsure. About 36% of all errors (1036 products) have a margin under 0.20.
- Errors happen between related categories. Book categories (used books, new books, book lots, magazines) make 7 of the top 10 confused pairs.
- Used Books (10) and Book Lots (2403) overlap because both use words like *livre*, *tome* and *édition*.
- Examples: *La Promenade Au Parc* (10) mentions *album* and *illustrations*, so it was predicted as Book Lots (2403). *Remington Bird Hunt Pour Wii + Fusil Noir* (40) mentions *jeu* and *fusil*, so it was predicted as Game Accessories (50).
- The next section checks if the image model does better on these categories.

8. Score per Category: Text vs Image

F1 for each category, for the text model (binary TF-IDF) and the image model (notebook 03_modeling_image). Both are scored on the same 16984 validation products.

```
Text validation products:  16984
Image validation products: 16984
Scored by both:            16984
```

Code	Text precision	Text recall	Text F1	Image precision	Image recall	Image F1	Products
10	0.523	0.543	0.533	0.519	0.281	0.365	623
1281	0.683	0.558	0.614	0.312	0.060	0.101	414
1180	0.814	0.601	0.692	0.301	0.144	0.195	153
2705	0.700	0.714	0.707	0.577	0.634	0.604	552
40	0.759	0.697	0.727	0.354	0.384	0.369	502
1280	0.737	0.719	0.728	0.315	0.352	0.333	974
2403	0.749	0.750	0.749	0.591	0.479	0.529	955
2582	0.788	0.734	0.760	0.324	0.425	0.367	518
2280	0.757	0.806	0.781	0.475	0.804	0.597	952
1140	0.794	0.796	0.795	0.465	0.573	0.513	534
2462	0.802	0.799	0.801	0.397	0.447	0.421	284
1302	0.830	0.801	0.815	0.316	0.157	0.209	498
1320	0.835	0.796	0.815	0.293	0.471	0.361	648
2060	0.811	0.833	0.822	0.526	0.308	0.389	999
50	0.797	0.851	0.823	0.692	0.054	0.099	336
2585	0.839	0.826	0.832	0.292	0.066	0.108	499
1560	0.842	0.855	0.848	0.468	0.514	0.490	1015
2220	0.929	0.794	0.856	0.579	0.067	0.120	165
60	0.944	0.819	0.877	0.471	0.675	0.554	166
1940	0.907	0.913	0.910	0.652	0.453	0.535	161
1920	0.901	0.927	0.914	0.683	0.760	0.719	861
1160	0.906	0.942	0.924	0.827	0.808	0.817	791
2522	0.929	0.942	0.935	0.484	0.570	0.524	998
1300	0.913	0.962	0.937	0.473	0.731	0.574	1009
1301	0.969	0.957	0.963	0.580	0.317	0.410	161
2583	0.970	0.985	0.977	0.668	0.666	0.667	2042
2905	0.983	0.983	0.983	0.545	0.385	0.451	174

Analysis

- Text: weighted F1 0.83. Image: weighted F1 0.48.
- Text is better on every category.
- Weakest text categories: Used Books (0.53), Board Games (0.61), Wargames (0.69), New Books (0.71).
- The image model comes closest to text on New Books (gap 0.10), Cards (0.11), Used Books (0.17) and Magazines (0.18). Book categories are where text is weakest, so combining both models may help there.
- The image model is close to useless for Game Accessories, Board Games, Garden Tools and Pets (image F1 about 0.10).

Conclusion

- Linear SVM with binary TF-IDF reaches 0.8313 weighted F1 and 0.8192 macro F1 on validation.
- Text cleaning does not change the score (0.8276 raw, 0.8274 clean).
- With 50000 features, the binary gain is within split noise. Using all features adds 1.2 points (CV on train).
- Most errors come from related categories that share words, mainly the book categories.
- Text beats image on every category. Images come closest on books.
- Next step (04): tune the model with cross-validation on train.